Cscope Tutorial and Tracing Mosquitto Programs

This is part of the course material for CSC0056 Data Communication, 2020Fall

Authors: 鍾尚值, 王超

NCL Networked Cyber-Physical Systems Laboratory

NATIONAL TAIWAN NORMAL UNIVERSITY

Outline

- What is cscope
- Cscope setup
- Using cscope to study Mosquitto programs
  - Publisher
  - Subscriber
  - Broker

What is cscope

- “ Cscope is a tool for developers to view source code. Cscope was part of the official AT&T Unix distribution for many years, and has been used to manage projects involving 20 million lines of code. “

http://cscope.sourceforge.net/

- Cscope support is built into Vim.

- There are many functions like finding symbols or finding all calls.

Cscope setup

- Use the following command to install the cscope.
  - *sudo apt-get install cscope*

- Generate cscope's database.
  - *cscope -q -R -b*

- Add the database to vim.
  - *:cs add cscope.out* (do this within vim)

-q	Build an inverted index for quick symbol searching.
-R	Recurse directories for files.
-b	Build the cross-reference only.

This way, however, we could only use cscope for programs in the same directory… (see the next page for improvement)

Cscope in a big project

- Find all program files in the root directory and create them in a cscope file.
  - *find . -name "*.c" -o -name "*.cpp" -o -name "*.h" > cscope.files*

- Based on file, generate the cscope's database.
  - *cscope -q -R -b -i cscope.files*

```
3MNN45C:~/mosquitto$ ls
     THANKS.txt        config.h        docker
d    about.html        config.mk       edl-v10
     aclfile.example   cscope.files    epl-v10
     appveyor.yml      cscope.in.out   examples
     buildtest.py      cscope.out      installer
pec  client            cscope.po.out   lib
     compiling.txt     doc             libmosquitto.pc.in
```

https://mark1990301.pixnet.net/blog/post/197749407-%E5%A6%82%E4%BD%95%E4%BD%BF%E7%94%A8cscope

Cscope in a big project (contd.)

- For the convenience of adding the database to VIM.
  - *cd /etc/vim*
  - *sudo vim vimrc*
  - Add the command at the end. *cs add ~/mosquitto/cscope.out ~/mosquitto* (as the line 58 shows)

```
55
56 set nu
57
58 cs add ~/mosquitto/cscope.out ~/mosquitto
59
60 set mouse=nv
```

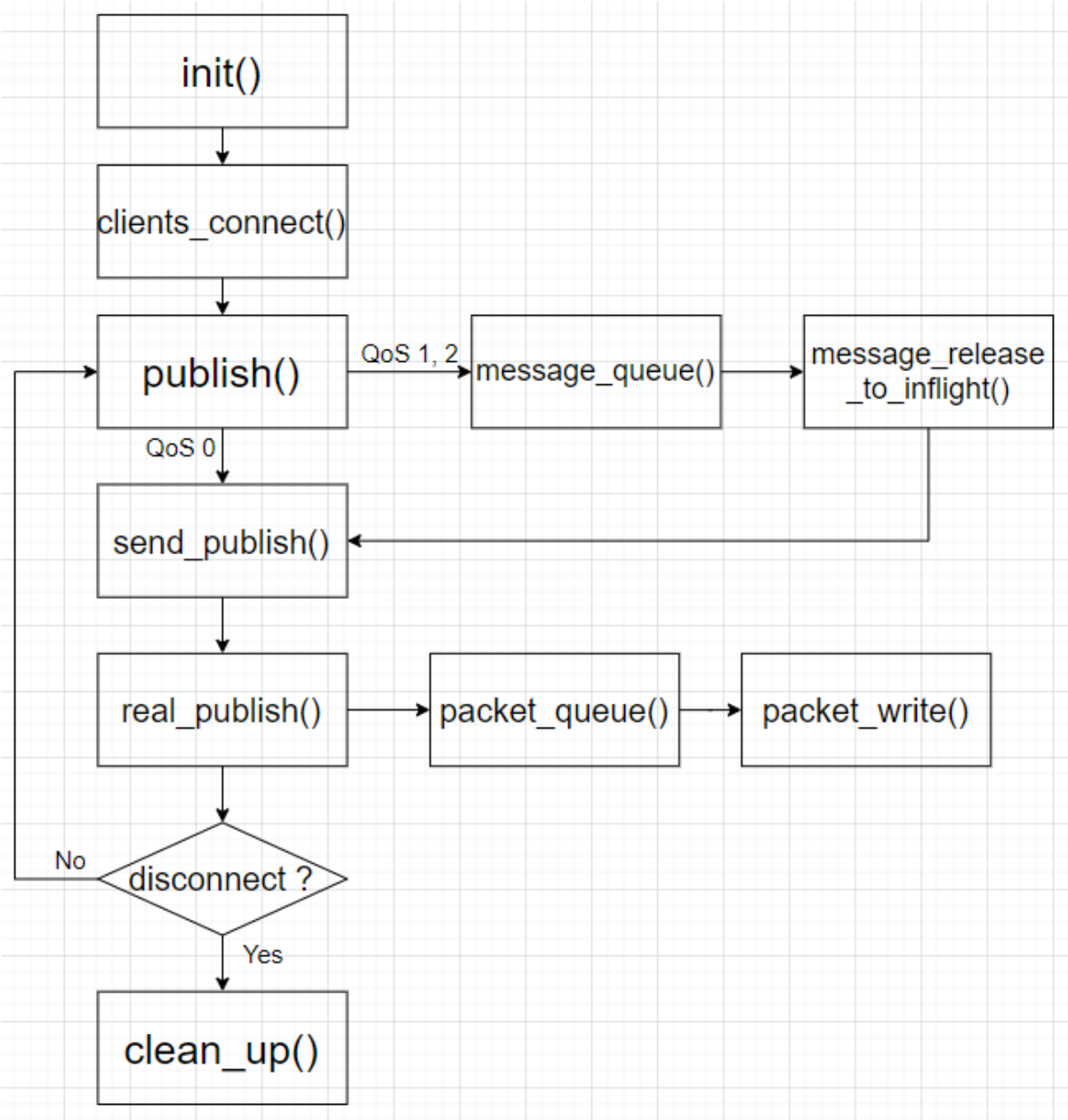
Cscope in a big project (contd.)

- When we use the cscope in vim.
  - *:cs find s|c|g name* And **s** is symbol, **c** is called, **g** is global definition.
- We may further speed up the process. Download cscope_maps.vim on the official website. And put it in the designated position.
  - *mkdir -p ~/.vim/plugin*
  - *cd ~/.vim/plugin*
  - *wget http://cscope.sourceforge.net/cscope_maps.vim*
- After completing all the above steps. We can use cscope more easily.
  - press "Ctrl + \" then s|c|g

https://cuteparrot.pixnet.net/blog/post/222677325-vim-%2B-cscope-%E9%81%94%E5%88%B0%E9%A1%9E%E4%BC%BC-source-insight-%E6%89%BE%E5%B0%8B-calls%2C-symbo

Example 1: Code tracing for Mosquitto publisher

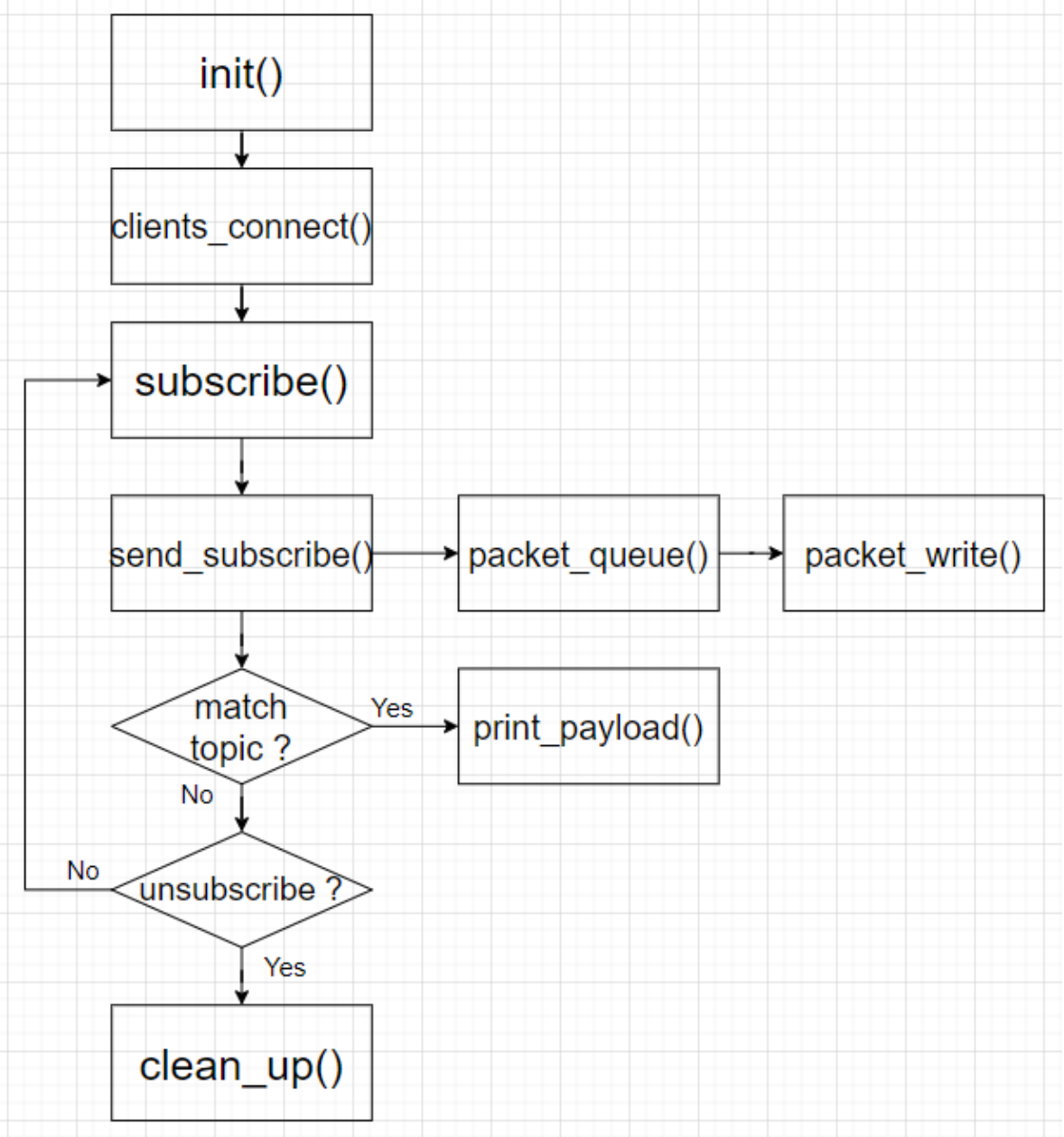
- send_publish(): set the transmission status to CMD_PUBLISH.
- real_publish(): send the data packet with transmission status to broker.
- If QoS > 1, the states will be set to wait_ack.



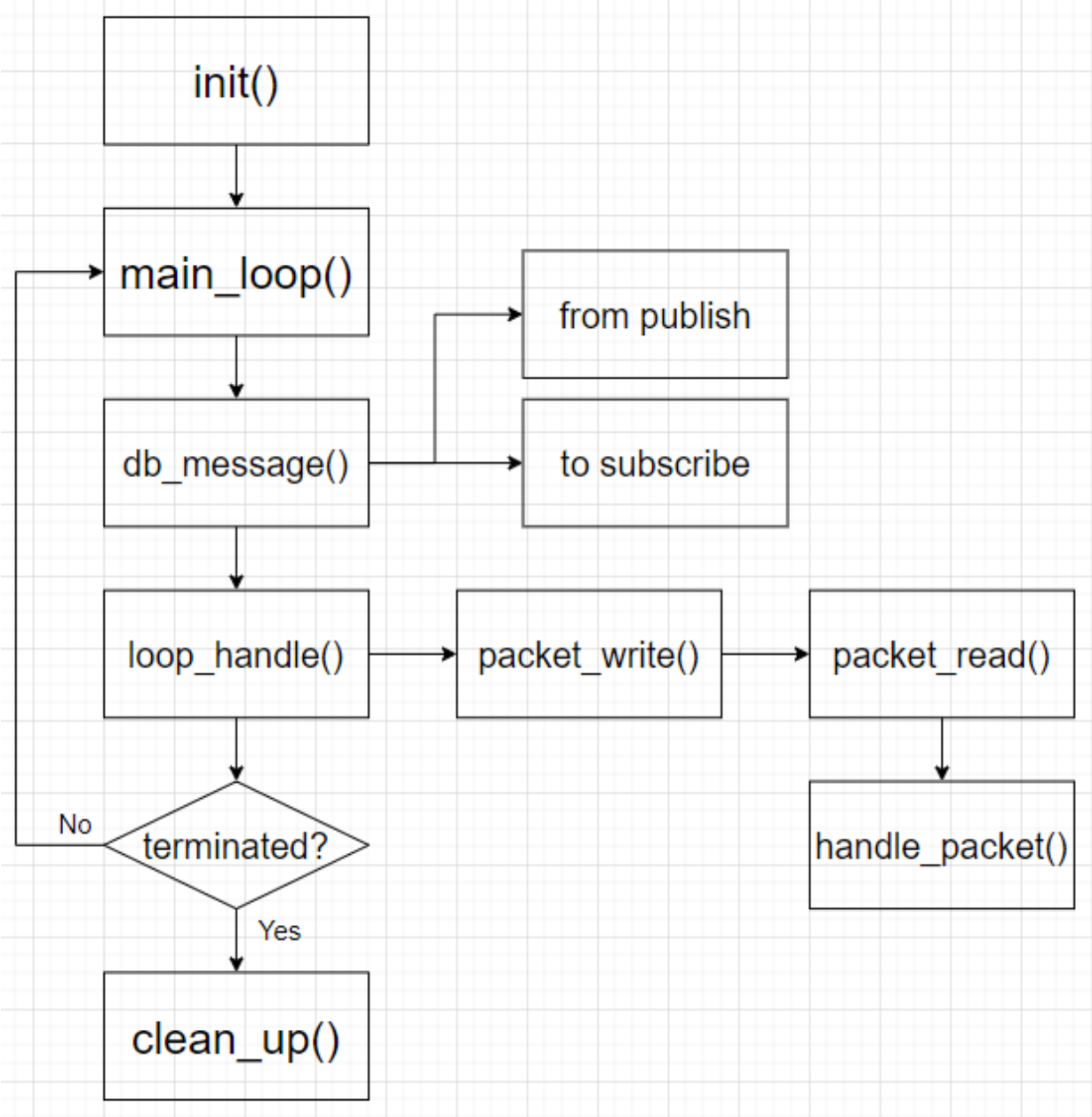
Example 2: Code tracing for Mosquitto subscriber

- send_subscribe(): set the transmission status to CMD_SUBSCRIBE and send to broker.

- If the subscription is successful, the content of the same topic will be displayed.



Example 3: Code tracing for Mosquitto broker

- db_message(): process control messages and set the transmission status such as CMD_PUBREC, CMD_PUBREL, CMD_PUBCOMP.

- handle_packet(): base on transmission status, It will send the corresponding message and call function to set next status CMD.